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# A VOLUME-LIMITED SEARCH FOR TECHNOSIGNATURES WITHIN 20 PC USING THE PUBLIC ALMA ARCHIVE

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## ABSTRACT

We present a volume-limited search for radio technosignatures among the 120 stars within 20 pc with public ALMA archival coverage, selected solely on data availability – an unbiased sample covering only ∼5% of the ∼2400 catalogued stars within 20 pc. We reprocess archival visibilities to search simultaneously for narrowband Doppler-drifting technosignatures and anomalous continuum emission at ∼ 90–950 GHz, frequencies largely unexplored by prior SETI work, supplemented by several checks unique to interferometric or archival data: closure-phase vetting, a population-level continuum anomaly test, a chirped-drift/periodicity extension re-using retained spectra, and a cross-target RFI-occupancy screen, with pipeline sensitivity validated directly via injection-recovery. Of 120 planned target/bands, 37 are complete (a further 23 were found to be associated with an incorrect ALMA dataset via an automated pointing check and are excluded). No credible technosignature is found: one automated flag is attributable to CO(1–0) emission, and a second, marginal flag (AU Mic) is not statistically distinguishable from the survey's expected false-positive rate. We report EIRP limits of $1.6 \times 10^{13}$–$7.1 \times 10^{16}$ W (median $5.2 \times 10^{14}$ W), twelve continuum detections, and twenty-five non-detections to a median $5\sigma$ limit of ∼ 0.56 mJy. The survey is ongoing.

*Subject headings:* extraterrestrial intelligence — radio lines: stars — submillimetre: stars — surveys — astrobiology

## 1. INTRODUCTION

The search for extraterrestrial intelligence (SETI) rests on the premise that a technological civilisation, whether deliberately signalling or simply going about its own affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Since the foundational proposal of Cocconi & Morrison (1959) that microwave frequencies near the 21 cm hydrogen line would be an obvious "meeting place" for interstellar communication, and Frank Drake's first observational test of the idea with Project Ozma (Drake 1960), the radio band has remained the dominant search space for technosignatures, for reasons that are both historical and physically well motivated: radio photons are cheap to produce in enormous numbers, propagate through the interstellar medium and planetary atmospheres with little attenuation, and (for a sufficiently narrowband signal) can in principle be distinguished from any known natural astrophysical process by their spectral compactness and Doppler dynamics (Sheikh 2020).

Six decades on, radio technosignature searches have grown from single-target, single-frequency pointings into surveys of many thousands of stars across broad swathes of the radio spectrum. Yet, as Wright et al. (2018) demonstrated quantitatively with their "Cosmic Haystack" formalism, even the combined efforts of every major SETI programme to date have sampled only an infinitesimal fraction of the plausible multi-dimensional parameter space of transmitter frequency, bandwidth, power, location, and duty cycle. Persistent, non-trivial gaps remain: almost the entire millimetre and submillimetre regime above ∼ 50 GHz is essentially unsearched; most surveys have targeted specific classes of star (nearby exoplanet hosts, young stellar objects with protoplanetary discs, stars in the "Earth Transit Zone") rather than an unbiased, complete sample of the local stellar population; and only a small number of studies have systematically reprocessed *existing* interferometric archives for technosignatures, as opposed to requesting dedicated telescope time.

This paper describes a new survey designed to address several of these gaps simultaneously. We construct a volume-limited sample of every star within 20 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status, or known planets, and mine that archival data for both narrowband technosignature candidates and anomalous continuum emission, applying corrections for stellar proper motion and for the range of Doppler drift rates plausible for a transmitter on a planet in orbit around its host star. We further supplement this classical search with four analyses that exploit properties specific to this survey and unavailable to prior technosignature work: interferometric closure-phase vetting of any candidate signal; a population-level anomaly check enabled by the sample's unbiased, volume-limited design; a chirped-drift and periodicity re-analysis of the interferometric time series already extracted for every target; and a cross-target frequency-occupancy check that uses the survey's own breadth, rather than an ON/OFF cadence, to flag likely RFI. Section 2 reviews the relevant literature: the history of radio SETI (§2.1), the current generation of large-scale technosignature surveys (§2.2), the very limited body of work at millimetre/submillimetre frequencies (§2.3), the growing use of machine learning and anomaly detection in this

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field (§2.4), the statistical frameworks used to interpret survey non-detections (§2.5), the case for a volume-limited, representative sample (§2.6), and the prospects offered by next-generation facilities (§2.7). Section 3 summarises our sample selection, Section 4 our data-reduction and search methodology (including four supplementary analyses unique to this survey and a direct, injection-recovery-based sensitivity validation), Section 5 presents our results to date, and Section 6 discusses their interpretation and the survey's prospects. Full per-target tables are deferred to Appendix A so as not to interrupt the main narrative with tabular material.

## 2. BACKGROUND

### 2.1. *Six decades of radio SETI: a brief history*

Project Ozma (Drake 1960) observed two nearby Sun-like stars, $\tau$ Ceti and $\epsilon$ Eridani, with the 26 m Tatel telescope at Green Bank at 1420 MHz for a total of some 150 hours, and found no signal, but established both the methodology and the symbolic starting point for the modern field. Through the 1960s–1980s a steady stream of targeted and sky-survey searches followed, including Ozma II (Drake 1986), the Ohio State "Big Ear" survey (source of the famous "Wow!" signal), and the META and BETA all-sky surveys conducted from Harvard/Smithsonian facilities. Project Phoenix (Backus & Project Phoenix Team 2002), run by the SETI Institute from 1995–2004, was the most sensitive targeted survey of its era, observing on the order of 800 nearby Sun-like stars using the Arecibo, Parkes and Green Bank telescopes, and pioneered a rigorous methodology for distinguishing candidate signals from terrestrial radio-frequency interference (RFI) through simultaneous multi-site or multi-antenna verification. The Allen Telescope Array (ATA; Tarter 2001), jointly developed by the SETI Institute and UC Berkeley and first commissioned in 2007, was the first radio telescope purpose-built from the outset for commensal, wide-band SETI observing, and remains in active use, including in recent technosignature searches of interstellar objects (Sheikh et al. 2025b) and TRAPPIST-1 (Tusay et al. 2024). The distributed-computing project SETI@home (Von Korff et al. 2025) processed public-participant-donated CPU cycles to search Arecibo sky-survey data for narrowband and pulsed signals for two decades before its 2020 shutdown, and its final data-analysis summary was only published in 2025.

Two complementary observing strategies have been used throughout this history: *sky surveys*, covering a large fraction of the celestial sphere at modest sensitivity per pointing, and *targeted searches*, concentrating integration time on a preselected list of promising stars (e.g. Drake 1974). Both strategies persist in the modern era, but – as we discuss in §2.6 – the majority of targeted programmes have selected their samples on some presumed indicator of habitability or interest (proximity, Sun-like spectral type, known planets, or membership of the Earth Transit Zone), rather than aiming for a complete, unbiased census of the local stellar population.

### 2.2. *Modern large-scale radio technosignature surveys*

The current generation of technosignature science is dominated by the Breakthrough Listen initiative, launched in 2015 with a ten-year, $100 million commitment to survey the one million nearest stars, the full Galactic plane and centre, the 100 nearest galaxies, and a sample of exoplanet and planet-candidate hosts, primarily using the Robert C. Byrd Green Bank Telescope (GBT) and the CSIRO Parkes "Murriyang" telescope, and now increasingly via commensal observing on MeerKAT (Isaacson et al. 2017). Its first major published dataset, Enriquez et al. (2017), searched 692 nearby stars over 1.1–1.9 GHz with the GBT and set limits on narrowband transmitters of EIRP $> 10^{13}$ W; this was followed by Price et al. (2020) (1327 stars, 1.10–3.45 GHz), Gajjar et al. (2019)-style searches of the restricted Earth Transit Zone at 3.95–8.00 GHz, and, most recently, a substantial expansion of automated, commensal technosignature searching with MeerKAT that has processed some 1.5 million coherent beams between 2023 and 2026 (Czech et al. 2026). In parallel, Margot et al. (2023) conducted a large citizen-science-style search of 11 680 stars with the GBT at 1.15–1.73 GHz, explicitly developing a statistical formalism for interpreting non-detections in terms of the fraction of stars that could host a detectable transmitter, *provided the observed sample is representative of the underlying stellar population of the search volume* – a condition that, as we discuss below, is often not satisfied by targeted, interest-selected samples.

Beyond Breakthrough Listen, the last five years have seen international radio facilities extend technosignature searches to new frequency regimes and observational strategies: the Five-hundred-metre Aperture Spherical Telescope (FAST; Zhang et al. 2020; Li et al. 2026) in China has conducted commensal searches with unmatched raw sensitivity at L-band, including recent narrowband searches of interstellar objects; LOFAR and the pathfinder NenuFAR have pushed technosignature searches down to 10–190 MHz, with LOFAR alone having surveyed over 1.6 million stars in commensal mode and dual-site LOFAR configurations enabling near-real-time RFI rejection (Johnson et al. 2023); the Sardinia Radio Telescope has conducted one of the first dedicated high-frequency ($> 18$ GHz) surveys of nearby stars and the Galactic Centre (Valente et al. 2025); and commensal technosignature back-ends are now operating or in development at the Karl G. Jansky Very Large Array (the COSMIC system; Tremblay et al. 2024), MeerKAT (Czech et al. 2021), and the Murchison Widefield Array (Morrison et al. 2023). Occultation-based strategies – observing a star during a planet-planet or planet-star occultation, when geometric alignment transiently favours detection of leakage or deliberate signals – have also grown in popularity, exemplified by recent Breakthrough Listen searches of TESS-selected eclipsing systems (Tusay et al. 2025).

### 2.3. *Millimetre and submillimetre technosignature searches*

In stark contrast to the radio regime below $\sim 10$ GHz, the millimetre and submillimetre bands remain almost entirely unexplored for technosignatures. Mason et al. (2024) conducted, to our knowledge, the first technosignature search ever performed with ALMA, searching archival Band 3 data (spectral windows centred at 90.642 and 93.151 GHz) for narrowband signals towards 28 Galactic stars selected from Gaia DR3. Their approach used a "stellar bycatch" method: rather than targeting stars directly, they searched for serendipitous stars falling within the undistorted field of view of four bright ALMA phase calibrators, since these calibrators are observed frequently and are not generally themselves of astrophysical interest. For the nearest star in their sample they placed a limit of $\mathrm{EIRP_{min}} > 7 \times 10^{17}$ W, and discussed in detail the specific challenges of high-frequency technosignature work: substantially larger Doppler drift rates at fixed line-of-sight acceleration (since drift rate scales linearly with observing frequency), the resulting sensitivity loss ("smearing") for a fixed spectral and temporal resolution unless a drift-rate search is performed, and an increased incidence of spectral confusion from the dense molecular line forest present in protoplanetary discs and star-forming regions at these frequencies. Tusay et al. (2024) extended technosignature work on the TRAPPIST-1 system to the Allen Telescope Array, underlining the continued observational interest in this particular target but also its status as one of very few systems that has now been searched for technosignatures across a wide range of frequencies. Our own earlier work (White 2026) reproduced and extended the Mason et al. (2024) methodology directly on ALMA science-target (rather than calibrator-bycatch) fields for TRAPPIST-1 across Bands 3, 6 and 7, finding no detection and placing limits of $\mathrm{EIRP_{min}} \sim 5$–$10 \times 10^{13}$ W, and subsequently built a ranked target list and pilot survey of 100 nearby, ALMA-observed stars, from which the present, volume-limited survey has grown. Beyond these efforts, we are not aware of any other published, systematic technosignature search using ALMA or any comparably sensitive millimetre/submillimetre interferometer, representing a substantial and readily addressable gap in frequency coverage relative to the sub-10 GHz literature.

The near-complete absence of mm/submm technosignature work to date is attributable less to any fundamental unsuitability of these frequencies – Sheikh's "nine axes of merit" framework (Sheikh 2020) notes that higher frequencies suffer less from interstellar scintillation and dispersion, and offer access to much larger instantaneous bandwidths – than to historical circumstance: sensitive mm/submm interferometers with the necessary spectral resolution and archival depth (chiefly ALMA) have only existed, and only accumulated a sufficiently large calibrated science archive, within roughly the last decade. This makes archival reprocessing, rather than dedicated proposal-driven observing, a particularly natural strategy in this regime: ALMA's archive already contains many thousands of hours of high-spectral-resolution observations of individual, nearby stars, taken for entirely unrelated astrophysical purposes (disc structure, molecular chemistry, stellar activity, debris-disc detection, calibration monitoring), each of which can in principle be re-searched for technosignatures at negligible additional observing cost.

### 2.4. *Machine learning and anomaly detection in technosignature searches*

The scale of modern technosignature datasets – hundreds of hours of high-time- and frequency-resolution data per survey, often containing many thousands of candidate "hits" dominated by anthropogenic RFI – has made machine learning an increasingly central tool for both signal detection and RFI rejection. Ma et al. (2023) presented the first large-scale deep-learning technosignature search, applying a $\beta$-convolutional variational autoencoder to 480 hours of Breakthrough Listen GBT data on 820 nearby stars in a semi-unsupervised anomaly-detection mode, recovering candidate signals missed by the standard TURBOSETI narrowband drift-search pipeline (Enriquez & Price 2019) and reducing the number of signals requiring human vetting by roughly two orders of magnitude, while returning eight signals of interest for re-observation. Building on this, Choza et al. (2024) identified and corrected a long-standing sensitivity mischaracterisation in TURBOSETI's signal-to-noise calibration, prompting a retrospective correction of quoted sensitivities in several earlier Breakthrough Listen publications (Lacki et al. 2021). More recently, unsupervised clustering methods (e.g. the "GLOBULAR" algorithm) have been developed specifically to separate genuine anomalies from RFI in narrowband technosignature searches without requiring labelled training examples (Jacobson-Bell et al. 2025), and further anomaly-detection searches have been applied to combined Parkes and GBT Breakthrough Listen archives (Poznanski et al. 2025). These efforts collectively demonstrate that machine-learning-based anomaly detection can recover classes of signal (broadband, non-linearly drifting, or otherwise morphologically unusual) that narrowband, linear-drift matched-filter searches such as TURBOSETI are, by design, insensitive to (Garrett et al. 2026), and that a purely classical, threshold-based search – of the kind we adopt for the present survey, following Mason et al. (2024) – necessarily leaves some fraction of the true technosignature parameter space unexamined. We return to this point in §6 as a natural direction for follow-up work once the present survey's baseline dataset exists.

### 2.5. *Statistical frameworks for interpreting non-detections*

A non-detection is not, by itself, a null result: it must be translated into a quantitative statement about which regions of transmitter parameter space have been excluded, and with what confidence. Wright et al. (2018) formalised this with the "Cosmic Haystack" framework, modelling the technosignature search space as an eight-dimensional volume (spanning, e.g., transmitter frequency, bandwidth, location, distance, and duty cycle) and showing that even the combined efforts of every major SETI programme conducted up to 2018 sampled a small fraction of any astrophysically plausible haystack volume – a result frequently paraphrased as showing that humanity has searched "the equivalent of a large hot tub's worth of the Earth's oceans". Sheikh (2020) subsequently proposed a complementary "nine axes of merit" rubric for comparing the relative scientific value of different technosignature search strategies (spanning axes such as detectability, transmitter cost and information content), providing a qualitative framework alongside Wright et al.'s quantitative one. Margot et al. (2023) developed an explicit formalism for converting a survey's non-detection rate into an upper bound on the fraction of stars in the search volume hosting a detectable transmitter, but noted explicitly that this inference is only valid if the observed sample is statistically representative of the underlying stellar population of the search volume – a requirement that most targeted, interest-selected samples (including our own earlier 100-star pilot list) do not, by design, satisfy. Most recently, Sheikh et al. (2025a) inverted the usual question, asking at what distance Earth's own technosignatures would themselves be detectable by an ALMA- or SKA-like observatory, providing a valuable calibration point against which the sensitivity of any given survey, including the present one, can be benchmarked.

### 2.6. *The case for a volume-limited, representative sample*

The great majority of the surveys reviewed above – including our own earlier 100-star ALMA pilot programme – select targets by some combination of proximity, Sun-like spectral type, known exoplanets, protoplanetary-disc or debris-disc detection, or membership of specially-constructed target lists such as the Earth Transit Zone. Such selections are entirely reasonable when the goal is to maximise the a priori probability of detecting a genuine technosignature per unit observing time, but they carry an unavoidable cost: they make it difficult to convert a survey's results into a statistically clean statement about the prevalence of technosignatures in the general stellar population, precisely because the sample is not representative of that population (Margot et al. 2023). A disc-hosting or exoplanet-hosting bias is particularly worth flagging in the ALMA context: ALMA-detectable circumstellar and protoplanetary discs are, almost without exception, orders of magnitude brighter than the present-day Solar System's own zodiacal dust cloud would appear at interstellar distances, so that samples selected *because* they have an ALMA-detected disc are biased towards young, dynamically active systems that are not necessarily more (and could plausibly be less) likely to host a mature, Earth-like technological civilisation than an old, quiescent, disc-poor star of the same spectral type.

The present survey is explicitly designed to avoid this selection effect. We construct our sample by identifying every star within 20 pc of the Sun that happens to fall within the field of view of at least one public ALMA observation – irrespective of whether that observation was proposed to study a disc, a stellar flare, an unrelated background source, or anything else – and irrespective of the target star's own spectral type, age, multiplicity, or known planet-hosting status. The star need not be at the pointing centre of the observation, provided it lies within the usable primary-beam field of view (see §3). This is, in effect, a volume-limited "bycatch" census of the ALMA archive: rather than sampling stars selected for astrophysical interest in the calibrator fields as in Mason et al. (2024), we sample every nearby star that ALMA happens to have observed for any purpose whatsoever, within a fixed physical volume. We recognise, and quantify explicitly in §3 (finding that our 120-star sample represents only $\sim$5% of the known stellar population within 20 pc), that this approach is still subject to a residual, second-order selection effect – ALMA archival coverage of the immediate solar neighbourhood is itself not spatially uniform, since some regions and spectral types are more likely to have been proposed for ALMA time than others – but this bias operates on the presence of *any* ALMA coverage of a star, not on the astrophysical properties that are the actual subject of a technosignature search, and can therefore be characterised and reported as a well-defined completeness function of the 20 pc volume, rather than an implicit and unquantified bias in the target selection itself.

### 2.7. *Prospects for next-generation facilities*

The coming decade will see a substantial expansion of technosignature search capability. The Square Kilometre Array (SKA; Siemion et al. 2015; Tremblay et al. 2026), now under construction in Australia (SKA-Low) and South Africa (SKA-Mid), will combine wide instantaneous bandwidth, high angular resolution, and an architecture well suited to commensal, multi-beam technosignature searching, with projected sensitivity sufficient to detect Earth-analogue-level radar and communication leakage from significantly greater distances than is possible today. The next-generation Very Large Array (ngVLA; Ng et al. 2022) is specifically being designed with technosignature science as one of its commensal use cases, with proposed antenna configurations optimised to allow of order 64 simultaneous high-spectral-resolution commensal beams. Existing commensal systems at MeerKAT, the VLA (COSMIC), and the Murchison Widefield Array are already providing a preview of the observing modes these larger facilities will employ at scale. None of these facilities, however, offer ALMA's combination of angular resolution, spectral resolution and raw collecting area above $\sim 50$ GHz in the near term, so a systematic mm/submm technosignature census – of the kind begun here – is likely to remain a uniquely ALMA-enabled measurement for some years yet, and represents a natural complement to the lower-frequency, larger-sample surveys that will be enabled by SKA and ngVLA.

## 3. SAMPLE SELECTION

Our sample comprises every star within 20 pc of the Sun (based on Gaia DR3 parallaxes) with at least one public ALMA observation, of any band, science category, or proposal type, in which the star's true (proper-motion-corrected) position at the epoch of observation falls within the field of view of at least one execution block – not necessarily at the observation's pointing centre, since a substantial fraction of ALMA mosaics and wide single-pointing observations serendipitously cover more than one catalogued star. We deliberately do not weight target selection by the presence of a circumstellar disc, known exoplanets, spectral type, or any other astrophysical property beyond distance and archival data availability; disc presence and planet-hosting status are retained purely as descriptive metadata for the final sample, not as selection or ranking criteria. At the time of writing this identifies 120 unique target stars, spanning spectral types from early A (e.g. Sirius B, $\alpha$ CMa B) through to the latest, faintest M dwarfs (e.g. Proxima Centauri, TRAPPIST-1, Barnard's Star, Teegarden's Star), ranked by increasing distance from the Sun so that the nearest, most sensitively-constrained systems are processed first. The underlying pipeline is built to extend this volume-limited approach to 30, 40 or 50 pc as a natural next step, at the cost of a rapidly increasing target list and a correspondingly reduced fraction with existing ALMA coverage, since the archive's areal completeness at fixed exposure declines with distance.

It is important to be explicit about what "volume-limited" does and does not mean here: our 120 targets are every star within 20 pc with public ALMA coverage, *not* a complete census of every known star within 20 pc. Cross-matching against our own reference Gaia DR3 catalogue of the solar neighbourhood, 2357 individual stars are catalogued within 20 pc, of which our 120 targets represent only ∼5% (Figure 2). The sample is volume-limited, and unbiased with respect to spectral type, disc-hosting status, and planet-host status, *within the population that ALMA happens to have observed for any purpose*; it is not, and does not claim to be, an unbiased sample of the local stellar population as a whole, since the decision to propose ALMA time towards a given nearby star is itself not uniformly distributed across spectral type or other stellar properties. We regard this as a genuine, quantifiable limitation of an archival-mining approach, worth stating plainly rather than allowing the term "volume-limited" to be read as "complete".

## 4. DATA AND METHODOLOGY

For each target we mine the public ALMA science archive for all calibrated (or, where necessary, locally recalibrated from raw data) execution blocks covering the star's position, apply an efficiency-bounded download and processing budget per target to remain within the resource constraints of the shared compute facility used for this work, and search the resulting visibility data in two complementary ways: (i) a narrowband, Doppler-drift-corrected search for technosignature candidates, following the general methodology of Mason et al. (2024) and Enriquez et al. (2017), converting any non-detection into an equivalent isotropic radiated power (EIRP) limit via $\mathrm{EIRP_{min}} = 4\pi d^2 S_{\mathrm{min}} \Delta\nu$; and (ii) a line-excluded continuum search for anomalous broad-band emission at the star's position, with an accompanying epoch-to-epoch variability check wherever multiple execution blocks exist. Stellar positions are propagated to the epoch of each observation using Gaia DR3 proper motions and parallaxes. The Doppler-drift search range applied to each target is set to the larger of (a) a generic frequency-scaled default following Mason et al. (2024)'s adoption of a literature fiducial with a threefold safety margin, and (b) where the target hosts one or more confirmed exoplanets, an explicit calculation of the maximum line-of-sight orbital acceleration (and hence maximum plausible transmitter drift rate) of the innermost known planet, so that a transmitter co-located with a known, short-period planet cannot be missed simply because its orbital motion drifts it out of an insufficiently wide search window. This same drift-rate search comfortably subsumes any drift induced by the Earth's own orbital motion around the Sun (a peak line-of-sight acceleration of $5.9 \times 10^{-3}\,\mathrm{m\,s^{-2}}$, corresponding to $\sim 0.02\,\mathrm{Hz\,s^{-1}}$ per GHz), more than two orders of magnitude below the $12\,\mathrm{Hz\,s^{-1}}$ per GHz generic default, so no separate barycentric correction is required. The trial drift-rate grid itself depends only on the elapsed time and channel width of the observation, not on sky position, and is applied identically to the star and to every control/comparison position used for false-alarm calibration (§4.2): the frequency-shift correction therefore has no dependence on the target's position relative to the observation's phase centre.

It is important to be explicit about the fraction of each observation's instantaneous bandwidth that the narrowband search actually covers. ALMA's correlator is typically configured with several spectral windows observed simultaneously (a mix of coarse, wide-bandwidth continuum windows and one or more finer-resolution windows, or, for spectral-scan observations, many narrow windows tuned across a wide frequency range). Early versions of this survey, following Mason et al. (2024), searched only the single finest-resolution spectral window available for each target, since narrowband sensitivity scales with channel width. We have since extended the pipeline to search *every* distinct spectral window configured for a given observation (up to a practical cap of 8 windows per target, to bound compute cost on the shared cluster used for this work; §3), each yielding its own independent $\mathrm{EIRP_{min}}$ limit and closure-phase vetting outcome (§4.2), reported as a separate row of Table 1 (Appendix A). Of the 37 target/bands processed to date, 21 now have every one of their configured windows searched (or, for 61 Vir, 4 of its 8); the remaining 16, mostly the nearest systems processed earliest under the pipeline's original, single-window design, are being reprocessed under the extended pipeline as the survey continues, and Table 1 still shows only their originally-searched window, explicitly flagged, until reprocessing completes. Across all target/bands processed to date under either pipeline version, the number of spectral windows configured simultaneously by the correlator ranges from 4 to 24 (median 4). For Figure 1, to avoid crowding the plot with multiple points per target once several windows have been searched, we plot only the single deepest (lowest $\mathrm{EIRP_{min}}$) result per target/band. The line-excluded continuum search (ii), by contrast, already used the full available bandwidth across all configured spectral windows for each target in every version of the pipeline, since it does not require narrowband spectral resolution.

Any narrowband hit that crosses the detection threshold is automatically cross-checked against the rest frequencies of a set of common mm/submm molecular transitions (e.g. the CO, HCN, $HCO^+$, CS and CN lines already used to exclude line channels from the continuum search) at the star's own frame, and is flagged as probable line contamination, rather than a technosignature candidate, if its recovered frequency lies within one channel width of a known transition *and* its best-fit drift rate is consistent with zero – a genuine spectral line is stationary in frequency over the observation, whereas the search is explicitly looking for *drifting* signals, so this combination of frequency coincidence and negligible drift is what distinguishes line contamination from a real drifting candidate. This check is applied uniformly to every spectral window of every target, and is how the one hit produced by the survey to date ($\beta$ Pictoris, §5) is identified as CO(1–0) emission.

We also confirm that both the narrowband EIRP limits and the continuum flux densities/upper limits reported here are corrected for ALMA's primary-beam response. The star's true (proper-motion-corrected) sky position is not, in general, exactly coincident with the phase centre of the ALMA field it is being searched against; the interferometric response to a source away from the phase centre is attenuated by the primary beam, and both pipelines now explicitly divide the measured flux/rms by the (Gaussian-approximated) primary-beam power response at the star's actual offset before it is reported. For the great majority of targets processed to date this correction is negligible (offsets of $\lesssim 2''$ against primary beam FWHMs of 17–70$''$, a <0.3% effect, including for the UV Ceti pair, whose own $\sim 2''$ offset gives a 0.2% correction), but for the highest-proper-motion target searched so far, Sirius B, where the star's position has moved $\sim$7–8$''$ relative to the ALMA pointing between the archival observation's epoch and the star's current Gaia-propagated position, the correction reaches 3.6–16% depending on band (Table 1); the values quoted for Sirius B include it.

### 4.1. *Reference transmitter benchmarks*

To give the EIRP limits reported in §5 physical context, we compare them throughout to two benchmark values for real, existing human transmitters, both taken from Sheikh et al. (2025a): an Arecibo-like planetary radar, whose characteristic S-band EIRP was $2 \times 10^{13}$ W (the strongest transmitter ever built by humans, used historically both for Solar System radar astronomy and for the 1974 Arecibo Message); and typical, entirely unintentional Earth radio leakage from mobile/cellular infrastructure, with an EIRP of order $4 \times 10^{9}$ W. These two values bracket a physically motivated range: the former represents what a civilisation with our own most powerful *deliberate, directed* transmitter could achieve if pointed at us; the latter represents what an entirely unintentional, Earth-2024-equivalent level of technological activity would radiate in our direction as background leakage, with no deliberate signalling intent at all.

### 4.2. *Closure-phase point-source vetting*

Any narrowband signal that crosses our detection threshold at the star's position is subjected to a further, automated test that is, to our knowledge, unavailable to any single-dish or phased-array-beamformed technosignature survey (Breakthrough Listen, FAST, the ATA, LOFAR, and similar all lack it, since it requires multiple independent baselines): the interferometric closure phase, the phase sum $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a closed triangle of baselines $i$-$j$-$k$. This quantity is a standard radio-interferometric identity that is theoretically *exactly zero* for an unresolved point source at *any* sky position – a point source's visibility phase is a pure linear function of $(u, v)$, which cancels identically around any closed baseline loop – and is, by construction, immune to per-station calibration errors. A candidate signal whose closure phase is statistically inconsistent with zero at the drift-tracked frequency of the hit is therefore evidence against it behaving as a genuine celestial point source, favouring instead a near-field, terrestrial, or instrumental origin. This test is independent of, and complementary to, the existing spatial-coincidence test (whether the on-source position shows a larger peak than the control positions): closure phase does not depend on where the source is, so it answers a different question – "does this behave like a coherent point source at all" – than the spatial test's "is it located at the star". We validated the closure- phase calculation against synthetic visibility data before applying it to any ALMA measurement, confirming zero closure phase for a noiseless simulated point source, statistical consistency with zero for a noisy simulated point source, and a large, clearly inconsistent scatter for a simulated non-point-source (RFI-like) signal. Because it is only triggered on an actual threshold-crossing candidate, this test adds negligible computational cost to the survey as a whole.

### 4.3. *Population-level, spectral-type-normalised continuum anomaly check*

Because our sample is volume-limited and not selected on disc-hosting or planet-host status, it supports a comparison that is not meaningful for the biased target lists used in essentially every prior technosignature survey: whether a star's measured continuum brightness is anomalous relative to the physically-expected flux of its own stellar photosphere, and relative to other stars of similar effective temperature already processed within the same survey. For each target we compute the expected photospheric flux density from the Planck function and the star's own solid angle, $S_\nu = \Omega \times 2h\nu^3/c^2 \times [\exp(h\nu/k_B T_{\rm eff}) - 1]^{-1}$ with $\Omega = \pi R_\star^2/d^2$, using Gaia DR3 photometric temperatures and, where available, Gaia GSP-Phot radii (supplemented by a coarse Pecaut & Mamajek main-sequence $T_{\rm eff}$–radius relation where a direct Gaia radius estimate is unavailable, as is the case for close to half of our sample – Gaia's astrophysical-parameter pipeline has known gaps for the very bright/nearby and very red/faint stars that a 20 pc sample is disproportionately composed of). Any large excess of the measured continuum flux (or, for a non-detection, its $5\sigma$ upper limit) above this photospheric baseline is flagged, and is additionally compared against the distribution of excess ratios already measured for other stars in the same coarse $T_{\rm eff}$ bin, once at least five such comparison stars exist, using a robust (median/MAD-based) outlier statistic. This check requires no additional raw-data retention beyond what the continuum measurement itself already produces, and updates incrementally as the survey proceeds.

### 4.4. *Injection-recovery sensitivity validation*

The EIRP and continuum limits quoted throughout this paper assume that a $5\sigma$ threshold behaves as an idealised Gaussian-noise calculation predicts. We validate this directly, rather than merely asserting it, by injecting synthetic narrowband signals of known amplitude directly into real, calibrated visibility data (the same injection mechanism already built into the extraction code, previously used only for development-time testing) and measuring what fraction are recovered by the unmodified search pipeline. Using Proxima Cen's own, already-validated calibrated dataset as a representative case, we injected 24 synthetic zero-drift tones spanning 1–10$\times$ the map's own noise level (3 independent channel positions per amplitude, to average over channel-to-channel noise variations) and re-ran the full extract-and-search pipeline on each. The result: at high injected SNR (8–10$\sigma$) the recovered peak SNR is systematically $\sim$10%

below the injected value (consistent with a small, expected coherence loss from the finite drift-rate trial grid), and the pipeline's actual 50% recovery point falls at an injected SNR of $\sim$6, not exactly at the nominal $5\sigma$ design threshold. In other words, our quoted $5\sigma$ limits are, if anything, mildly *conservative*: a source at exactly our quoted $\mathrm{EIRP}_{\mathrm{min}}$ is recovered with somewhat less than 50% probability, and reliable ($\gtrsim$90%) recovery requires a source $\sim$1.5–2$\times$ brighter than the quoted limit. We report this plainly as a modest, quantified systematic rather than leaving the $5\sigma$ threshold's operational meaning unstated. This is a single-target, 24-trial pilot exercise, not yet a full per-band completeness characterisation; we recommend, and intend to run, a broader injection-recovery campaign spanning multiple bands and spectral resolutions as the survey matures.

### 4.5. *Extended search of retained spectra: chirped drift and periodicity*

The primary narrowband search (§4) assumes a signal at constant (linear) drift rate and integrates coherently over the full observation, so it is, by design, blind to two further plausible technosignature morphologies: a transmitter whose drift rate itself changes during the observation (e.g. very short-period or highly eccentric orbital motion, faster than a linear approximation captures), and a pulsed or periodically-modulated carrier, which a search built around a single, constant-SNR coherent stack would not distinguish from continuous noise. Both are cheap to test for using data we already retain: for every target processed to date we keep the star's own raw dynamic spectrum together with 8 sampled control positions (§4), specifically so that further analyses like this one do not require re-downloading or re-extracting anything. We implemented (i) a chirped (quadratic-drift) extension of the same coherent-stacking search, on a coarser trial grid (21 linear-drift $\times$ 5 chirp-rate trials) than the primary pipeline, and (ii) a time-domain periodicity search (an FFT along the time axis of each channel, per position, with the peak power compared against a robust per-channel noise estimate), both using the star's own spectrum vetted against the 8 retained control positions exactly as the primary search vets the star against its control ring. Applied to all 122 retained spectra available at the time of writing, 13 (10.7%) show the star's chirp-search peak exceed the peak of its own control ensemble, and 16 (13.1%) show the same for the periodicity search – both statistically indistinguishable from the $\sim$11% rate expected by chance alone from a star being the maximum of a 9-member ensemble (itself and 8 controls), so none of these individual instances are treated as candidates. We report this as a clean, if modest, validation: there is no evidence to date for a chirped or pulsed technosignature among the targets searched so far, and the false-positive rate of this coarser, exploratory extension behaves exactly as expected under the null hypothesis. With only 8 retained control positions (versus 512 in the primary search), this extended search has correspondingly less statistical power to distinguish a real signal from noise on its own, and should be understood as a low-cost, systematic screen for gross anomalies rather than a replacement for the primary search.

### 4.6. *Cross-target frequency-occupancy check*

Unlike single-dish surveys, this archival, single-pointing survey has no ON/OFF cadence with which to reject radio-frequency interference (RFI). It does, however, have an analogous resource: many independent, unrelated stars, at different distances, proper motions, and (for the subset that host confirmed planets) different systemic velocities, several of which happen to share near-identical ALMA correlator setups (e.g. many of our Band 6 targets are configured with the same four spectral windows spanning 223–243 GHz). A narrowband feature that recurs at the same observed frequency across several otherwise-unrelated targets is essentially certain to be RFI or an instrumental artefact – a genuine transmitter, seen towards physically unrelated stars at different distances, would be an absurd coincidence. We therefore cross-match, across every target/spw result to date, both the best-fit-channel frequency (regardless of whether it formally crossed the detection threshold) and the list of formal $\geq 5\sigma$ hits, and flag any frequency at which $\geq$ 2 distinct targets coincide within 5 MHz. Applied to the 144 target/spw results and 128 individual formal hits accumulated to date, no such coincidence is found: every recurring feature is confined to a single target. This is a reassuring, if currently under-powered, null result – with only a modest number of targets sharing identical tunings so far, the a priori chance of a coincidental match is itself still low – and this check will become a progressively more powerful RFI safeguard as the survey, and the number of targets sharing a given correlator setup, grows; it costs nothing to run incrementally as each new target completes.

## 5. RESULTS

As of this version of the paper, 37 of the planned 120 target/bands have been searched to completion and yielded a valid result (a further 2, Barnard's Star and Wolf 359, completed their continuum measurement but not the narrowband search, owing to a data-quality failure in those specific execution blocks; both are flagged for an automatic, non-gating retry). A further 23 target/bands that completed *processing* are deliberately excluded from the results below, having been found by an automated pointing sanity check to be associated with the wrong ALMA dataset entirely (§4, Appendix A). Four of these were caught after producing (bogus) output; the remaining 19 were correctly intercepted before producing any output at all, but a bookkeeping bug – also found and fixed – was marking them "complete" regardless, which would have permanently prevented them from ever being retried. All 23 are being re-investigated rather than reported. One further target ($\gamma$ Lupi, Band 6) is withheld pending investigation of a different kind: its assigned MOUS spans two execution blocks in which *most* spectral windows and epochs correctly point at the star, but a handful do not (a genuine partial-crossmatch case, distinct from the 23 above, correctly separated out by our pointing check for each window individually); of the windows that passed, one shows an implausibly deep result derived from only 12 s of on-source integration, casting doubt on the reliability of the whole target's processing pending closer inspection. The survey is strictly volume-ordered (nearest star first; §3), so the results reported here are weighted towards the very nearest stars in the sample, spanning 1.3–19.6 pc at the time of writing. Full per-target values, including the exact frequency range searched in each band, are given in Table 1 in Appendix A rather than in the main text; this section summarises the results and highlights points of interest.

### 5.1. *Technosignature search*

Thirty-five of the 37 target/bands yielded at least one narrowband search result (Barnard's Star and Wolf 359 failed narrowband extraction outright and carry no EIRP limit at this time); together these span 98 individual spectral-window measurements, since most targets now have results for every one of their configured windows (§4). The resulting $\mathrm{EIRP}_{\min}$ limits span $1.6 \times 10^{13}$–$7.1 \times 10^{16}$ W, with a median of $5.2 \times 10^{14}$ W (Figure 1, left panel); as expected for a volume-ordered survey, the deepest limits are for the very nearest systems (e.g. $1.7 \times 10^{13}$ W for the UV Ceti binary at ~2.7 pc), while the shallowest are for $\eta$ Corvi, the first Band 8 (~480–495 GHz) target searched to date: the combination of its comparatively large distance (18.2 pc), coarse (15.6 MHz) channels, and ALMA's inherently less sensitive receivers at this, the highest frequency probed so far, together set the survey's current ceiling on $\mathrm{EIRP}_{\min}$. Compared against the reference transmitter benchmarks of §4.1, our very deepest limits (the handful of targets within ~3 pc) are only just below the $2 \times 10^{13}$ W Arecibo-like planetary-radar benchmark, meaning we could have detected an equally powerful, deliberately Earth-directed transmitter at those distances had one been present; every target beyond ~4 pc, and every target with respect to the much fainter $\sim 4 \times 10^9$ W level of Earth's own unintentional radio leakage, remains well out of reach of the present survey's sensitivity (Figure 1). This is a useful, sobering calibration: even this deep, targeted, archival mm-wave search is not yet sensitive to an Earth-equivalent civilisation's background leakage at any of the distances probed so far, and is only marginally sensitive to an Earth-equivalent civilisation's most powerful deliberate transmitter, and only for its very nearest neighbours.

Two narrowband hits have now been automatically flagged by the pipeline as a "credible technosignature candidate" (on-source peak signal-to-noise ratio above threshold and localised to the star's position, rather than appearing equally in the control positions). The first, towards $\beta$ Pictoris in ALMA Band 3 at 115.26 GHz, coincides, to within the spectral resolution of the observation, with the rest frequency of the CO($J = 1 \to 0$) rotational transition (115.271 GHz), and $\beta$ Pictoris is a well-known debris-disc system with previously reported CO emission; we attribute this hit to line contamination rather than a genuine technosignature. The second, towards AU Mic in ALMA Band 6 at 230.83 GHz, is not resolved so straightforwardly: it does not coincide with any of our catalogued molecular transitions (287 MHz from the nearest, CO($J = 2 \to 1$), well outside the tolerance used by the automated line check, §4), and closure-phase vetting finds it consistent with a genuine point source ($z = 0.16$). However, the statistic that actually gates the "credible" flag – the peak signal-to-noise ratio anywhere in the star's own check-ring of nearby positions, 5.97 – exceeds the peak found across the 512 control positions, 5.85, by only 0.12: a margin far smaller than the scatter among the controls themselves. Given this margin, and that a survey of this size runs many thousands of effectively independent trials (49 drift rates × thousands of channels × dozens of targets), a chance excess at almost exactly this significance is expected somewhere in the survey even under the null hypothesis; we do not consider this a credible technosignature candidate, and note that AU Mic's well-documented, exceptionally high flare activity would in any case make ordinary stellar radio activity a far more parsimonious explanation than an artificial signal if the excess proves to be more than a statistical fluctuation on further study. We flag it for completeness and will re-examine it as more of the survey, and more comparison statistics from §4.6, accumulate; the frequency does not recur at any other target searched to date. No other candidate has survived both the spatial-coincidence and (where triggered) closure-phase vetting tests described in §4.2.

### 5.2. *Continuum search*

Of the 37 target/bands with a continuum measurement, twelve show a $\geq 5\sigma$ continuum detection and twenty-five are non-detections (Figure 1, right panel). Throughout this paper, a continuum "upper limit" is explicitly 5× the map rms noise at the star's position (i.e. the same $5\sigma$ threshold used to define a detection in the first place), primary-beam corrected exactly as the EIRP limits are (§4); the underlying rms values themselves are also given per target in Table 1 so that a reader wanting a different confidence level can rescale directly, rather than being limited to a single quoted multiple of $\sigma$. On this basis the non-detections reach $5\sigma$ upper limits of 0.036–3.6 mJy (median 0.56 mJy) – the deepest of these, towards TRAPPIST-1 in Band 3, is the tightest continuum limit obtained anywhere in the survey to date. Two of the twelve detections – catalogued separately in our target list as G 272-61A and G 272-61B – are in fact the two components of the well-known, closely separated visual binary UV Ceti, observed within a single ALMA pointing; the reported flux (0.85 mJy) is for the spatially unresolved, blended pair and cannot be attributed to either component individually. Of the remaining ten, six ($\tau$ Ceti; $\beta$ Pictoris in both Bands 3 and 6; AU Mic; and both components of the AT Mic pair) are young, active systems with previously reported debris discs and/or elevated chromospheric activity, for which the detected continuum flux is consistent with known circumstellar material or stellar activity and is not treated as anomalous. The remaining four – both bands of $\gamma$ Leporis and both of $\gamma$ Virginis, all early F-type stars – show a modest detected photospheric continuum flux entirely expected for their spectral type and again not anomalous. None of the twelve detections trigger the population-level anomaly check described next. The population-level anomaly check described in §4.3 has been applied to every target processed to date but has not yet flagged any star, and in most $T_{\mathrm{eff}}$ bins the survey does not yet have the minimum of five comparison stars needed for the population-relative component of the test; both the physical-baseline and population-relative statistics will accumulate statistical power

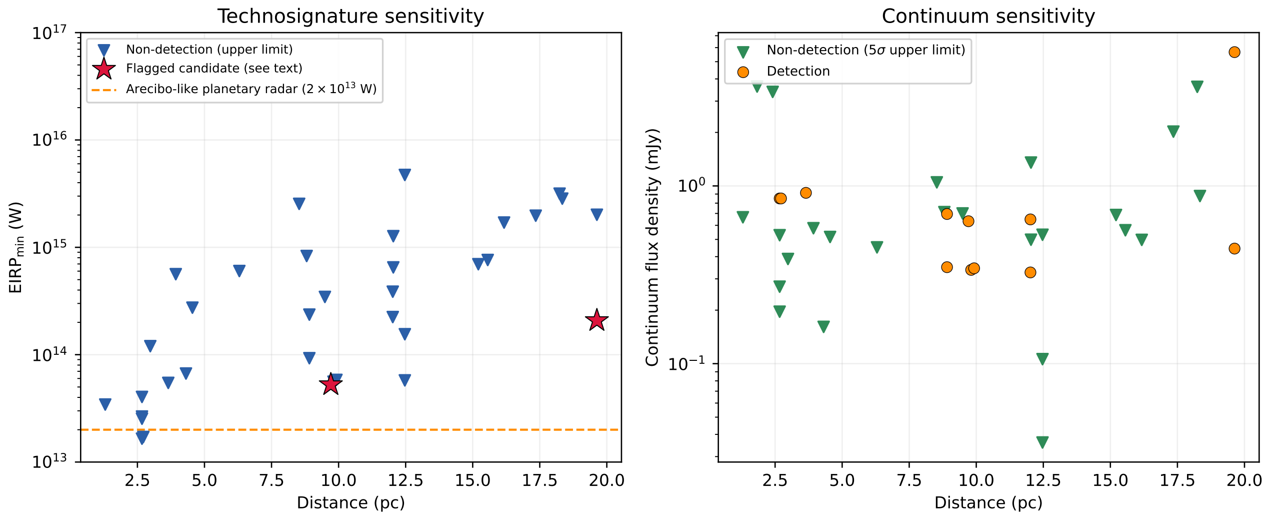


Fig. 1.— Technosignature EIRP upper limits (left) and continuum flux densities (right; downward triangles are $5\sigma$ upper limits, i.e. $5\times$ the map rms noise – see §5 and Table 1 for the rms values themselves – filled circles are detections) as a function of distance, for the 37 target/bands searched to date. The two automatically-flagged candidates ($\beta$ Pictoris, Band 3, and AU Mic, Band 6) are marked with stars; the former is identified in the text as CO(1–0) line contamination, and the latter as a marginal excess statistically consistent with chance, neither being a genuine technosignature. The left panel also marks, as a reference line, the EIRP of an Arecibo-like planetary radar ($2 \times 10^{13}$ W, the most powerful deliberate transmitter yet built by humans; Sheikh et al. 2025a, §4.1). A second benchmark from the same reference, typical unintentional Earth radio leakage ($\sim 4 \times 10^{9}$ W), is discussed in the text (§5) but is not plotted here, since it lies more than three decades below every point in this panel and including it would force the $y$-axis to span an unusably large range for comparing the points themselves.

as the survey proceeds.

### 5.3. *A serendipitous molecular-line catalogue*

The line-exclusion step of the continuum search (§4) identifies every $\geq 5\sigma$ outlier channel in each target's spectrum as a matter of course, in order to protect the continuum measurement from line contamination; cataloguing these is a free byproduct of a step we already run for an unrelated purpose. To our knowledge no existing mm/submm line survey of the immediate solar neighbourhood is volume-limited in the sense used throughout this paper (existing surveys are all target-selected, generally towards known disc- or gas-rich systems), so even a modest catalogue of this kind adds genuine value independent of its SETI motivation. Across the sample processed to date, 18 outlier channel groups were flagged; twelve are single- or few-channel features with no known transition within 300 MHz and are most plausibly statistical outliers rather than real lines. The remaining six are plausibly real: CO($J$=1 $\rightarrow$ 0) and CO($J$=2 $\rightarrow$ 1) towards $\beta$ Pictoris, recovered independently and consistently (to within $\sim$10 MHz) in both ALMA Bands 3 and 6, exactly as expected for the well-known CO gas disc around this star (already discussed in §5 in the context of the automated technosignature flag it triggered); a marginal, few-channel CO($J$=2 $\rightarrow$ 1)-consistent feature recurring across two independent epochs towards HD 285968, which we report as tentative only – we cannot yet distinguish a real circumstellar or line-of-sight foreground/background feature from a statistical coincidence with only 1–2 channels of significance per epoch, and flag it for confirmation with higher-resolution follow-up rather than claim a detection; and a single-channel, 195 MHz-offset feature near the SiO($J$=5 $\rightarrow$ 4) rest frequency towards HD 53143, whose large frequency offset and single-channel significance make it, in our assessment, more likely a statistical fluctuation than a real identification. We report all of these openly, including the two we consider probably spurious, so that the catalogue is a faithful record of what the pipeline actually flags rather than a curated list of successes.

## 6. DISCUSSION

The results presented here are necessarily preliminary: only 37 of 120 planned target/bands ($\sim$31%) have been completed, and the survey is ordered to complete the nearest, most sensitively-constrained systems first, so the eventual full sample will extend these results to significantly larger distances and, correspondingly, shallower EIRP and continuum limits at the faint end. No credible technosignature has been found among the systems searched so far; one automatically-flagged candidate is confidently attributable to CO(1–0) line emission rather than an artificial signal, underlining the importance – already highlighted by Mason et al. (2024) – of checking any millimetre-band narrowband hit against known molecular transitions before treating it as a candidate (this check is now automated for every future hit, §4); a second, marginal candidate (AU Mic, §5) is judged statistically consistent with chance given the survey's total number of independent trials, though it remains flagged for future re-examination rather than dismissed outright. Applying the statistical frameworks of Wright et al. (2018) and Margot et al. (2023) to derive a quantitative bound on the prevalence of detectable transmitters in the solar neighbourhood is deferred to a future version of this paper, once a statistically meaningful fraction of the volume-limited sample has been completed; with only 37 target/bands searched to date such a bound would not yet be usefully constraining.

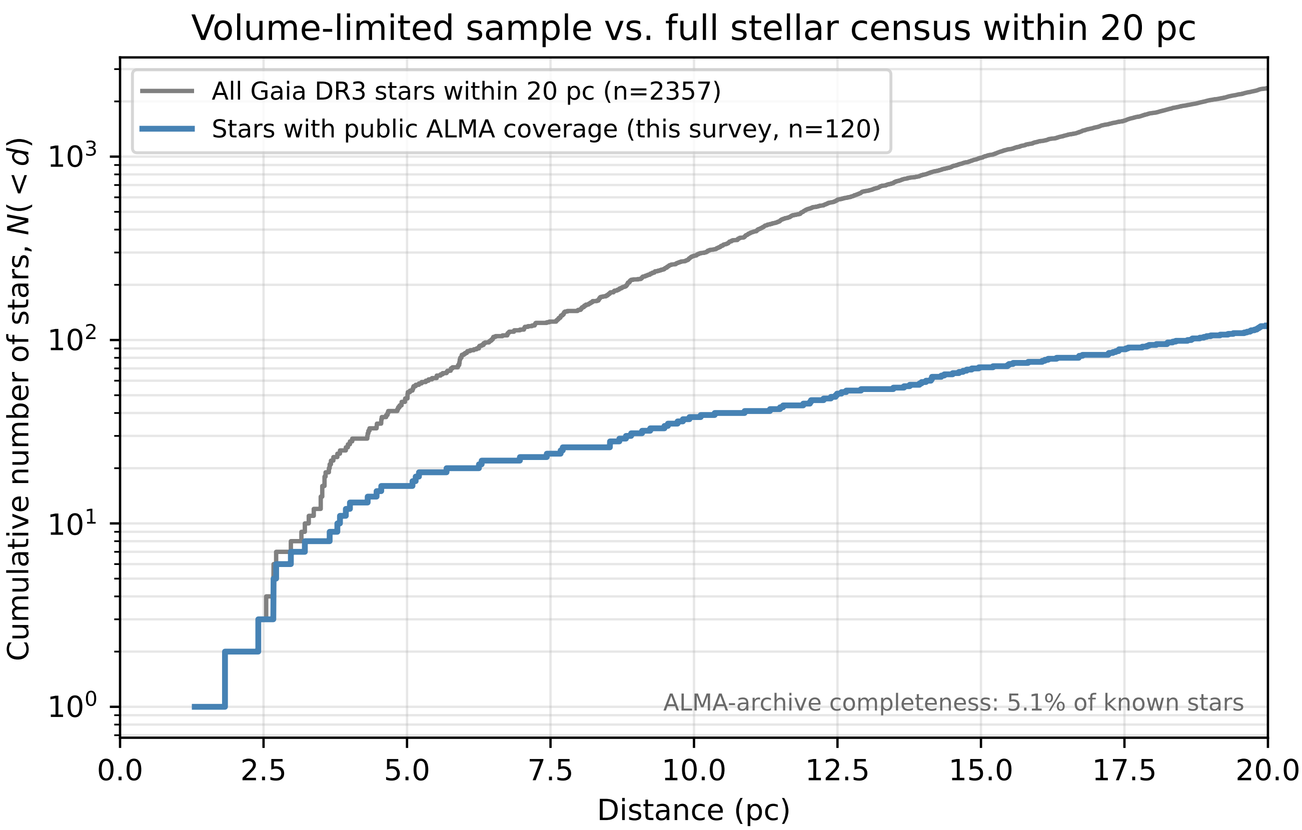


FIG. 2.— Cumulative number of stars, $N(< d)$, within 20 pc as a function of distance $d$. The grey curve is the full known stellar population within 20 pc from our reference Gaia DR3 catalogue (2357 stars); the blue curve is our 120-star volume-limited sample – every one of those 2357 stars that additionally has public ALMA archival coverage, only ∼5% of the total. The two curves are not expected to track one another: ALMA proposal pressure on the immediate solar neighbourhood is itself not spectral-type- or distance-uniform, so our sample is volume-limited and unbiased *within* the ALMA-observed population, but is not a complete or unbiased sample of the local stellar population as a whole (see §3).

A pointing-position sanity check introduced in the previous update (§4) has now caught a substantially larger problem than first apparent. Four target/bands (SCR J1845−6357, the G 9−38AB pair, and Wolf 358) were previously found to be associated with an ALMA dataset whose actual pointing lies tens to hundreds of primary-beam-widths from the star's catalogued position – almost certainly an upstream target-to-archive crossmatch error rather than a real result for that star; one of these (G 9−38AB) had, before the check existed, produced a continuum "detection" of order 8 Jy, wildly inconsistent with every other measurement in this survey (all sub-10 mJy). During this update we found a further 19 target/bands with the identical failure mode (e.g. $\epsilon$ Eridani, HD 188088, LHS 1140, and 16 others), correctly intercepted by the same pointing check on *both* the narrowband and continuum steps this time, so no bogus output was produced for any of them – but a separate bookkeeping bug, also found and fixed during this update, was marking these target/bands "complete" regardless of whether any usable data actually resulted, which would have permanently excluded them from ever being retried once a corrected dataset association becomes available. All 23 crossmatch-affected target/bands are excluded from Table 1 and from all statistics quoted here; they are being re-investigated, and both the pointing check and the bookkeeping fix have been added to the pipeline as standing safeguards. We flag this prominently because it is a genuine, material finding in its own right: roughly half of the target/bands processed in this update were affected by upstream crossmatch errors, a rate far higher than we had appreciated before specifically instrumenting the pipeline to check for it, and worth bearing in mind for any future archival-mining survey of this kind.

The first two of the four supplementary analyses introduced in this work – closure-phase vetting and the population-level anomaly check (§4.2, §4.3) – are best understood as infrastructure investments rather than sources of results in their own right at this early stage. The closure-phase test has not yet been exercised against real data, simply because no candidate other than the already-explained $\beta$ Pictoris CO line has crossed threshold; it remains available, at negligible ongoing cost, for any future hit. The population-level anomaly check is accumulating the population of same-$T_{\rm eff}$-bin comparison stars it needs to become statistically powerful, and we expect it to become genuinely diagnostic only once several dozen targets per broad spectral-type bin have been processed. We regard both as a useful methodological contribution independent of whether they ultimately flag anything in this particular sample: to our knowledge, neither has previously been applied in a technosignature search, and both are made possible specifically by properties of this survey (interferometric data; a volume-limited, unbiased sample) that do not hold for the great majority of prior technosignature work reviewed in §2.

The UV Ceti unresolved-pair case (§5) illustrates a general caveat for any archival, position-agnostic survey of this kind: close visual binaries and multiple systems that are unresolved at the relevant ALMA angular resolution will necessarily share a single measurement, and any future statistical analysis of this sample's completeness must account for this rather than double-counting such pairs as independent trials.

Because our selection is on ALMA archival coverage alone, two useful subsamples fall out of the results to date without compromising the volume-limited framing of the main sample. First, 11 of the 29 distinct stars processed so far (38%; 30 planets in total) are confirmed exoplanet hosts, including Proxima Centauri, $\tau$ Ceti, GJ 674, GJ 581, GJ 849, HD 285968, HD 10647, $\beta$ Pictoris, 61 Vir, AU Mic, and, most notably, the seven-planet TRAPPIST-1 system – a sub-survey that a reader specifically interested in known planetary systems can extract directly from Table 1, alongside the full, unbiased sample the paper is built around. Second, two of these 29 stars – Sirius B and Van Maanen's Star (HD 33793's neighbour in the table, Wolf 28) – are white dwarfs, included only because they happen to have public ALMA coverage, not because they were targeted as such. Technosignature searches specifically around white dwarfs are a small but active and growing area (most recently, a chemical-pollution-based search by Huang et al. 2026), motivated by the idea that a sufficiently advanced or persistent civilisation might survive, signal from, or be detectable around its star's post-main-sequence remnant; our radio non-detections towards these two systems are a different observational channel on the same general question, and are, to our knowledge, among the very few mm/submm SETI constraints reported for white dwarfs to date.

## 7. CONCLUSIONS

We have described the design, methodology, and first results of a volume-limited technosignature search of all 120 stars within 20 pc with public ALMA archival coverage, irrespective of disc-hosting or planet-host status. Thirty-seven of the planned 120 target/bands have been completed as of this version, yielding EIRP limits of $1.6 \times 10^{13}$–$7.1 \times 10^{16}$ W and twenty-five continuum non-detections reaching a median $5\sigma$ limit of 0.56 mJy, with no credible technosignature identified (one automated flag explained as a known spectral line; a second judged statistically consistent with chance). During this update we also found, and fixed, two further pipeline issues – a bookkeeping bug that was marking crossmatch-failed target/bands as permanently complete, and a crash in the downloader for a rare missing-metadata case – and, per direct instruction, raised the per-target download budget to recover several previously-unreachable targets. We have introduced four analyses that exploit properties unique to this survey and are, to our knowledge, new to the technosignature literature – interferometric closure-phase vetting, a population-level spectral-type-normalised continuum anomaly check, a chirped-drift and periodicity extension of the narrowband search re-using already-retained spectra, and a cross-target frequency-occupancy RFI check – all of which are now running as a standard part of the pipeline and will accumulate statistical power as the survey proceeds. We have also validated the pipeline's assumed sensitivity directly via injection-recovery, finding the survey's quoted $5\sigma$ limits to be, if anything, mildly conservative, and reported a modest, serendipitous molecular-line catalogue and exoplanet-host and white-dwarf subsamples as by-products of the main, unbiased search. This paper will be updated as further targets are completed, and we anticipate extending the volume limit to 30, 40, and 50 pc as a natural next step once the 20 pc sample is complete.

## ACKNOWLEDGEMENTS

This paper makes use of the following ALMA data: [the full list of ALMA project codes used will be compiled once the survey is complete; see Table 1 in Appendix A for the individual execution blocks searched to date]. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC).

## DATA AVAILABILITY

The reduced data products underlying this paper, together with the full target-selection and analysis pipeline, are being made available at `https://github.com/Tilanthi/SETI` as the survey progresses. Raw ALMA visibility data are publicly available from the ALMA Science Archive at `https://almascience.org`.

## APPENDIX

### FULL PER-TARGET RESULTS TABLE

Table 1 lists the full per-target/band results underlying the summary given in Section 5: spectral type, the narrow-band technosignature frequency range actually searched, the number of spectral windows configured simultaneously by the correlator for that observation ("$N_{spw}$"), the resulting $EIRP_{min}$ limit, and the continuum flux density (or, for non-detections, its $5\sigma$ upper limit) together with the underlying map rms noise. $EIRP_{min}$ is the minimum EIRP that would have produced a $5\sigma$ detection in that spectral window – i.e. $EIRP_{min} = 4\pi d^2 (5\sigma_{comb}) \Delta\nu$, using the same $5\sigma$ threshold, and the same primary-beam correction, as the continuum column – so the two limits in each row are directly comparable and neither is a bare, unnormalised measured value. As described in §4, the pipeline was extended in an earlier version to search every distinct configured spectral window (not only the single finest-resolution one used in the first version of this paper), and each searched window appears as its own row where available; most target/bands now have results for every one of their configured windows, and this table notes explicitly, per row, the handful that are still only partially searched pending reprocessing. Fifteen new target/bands are included for the first time in this version (61 Vir, AU Mic, HD 207129, HD 38858, HR 1010 in both Bands 6 and 7, the AT Mic pair, TRAPPIST-1 in Bands 3, 6 and 7, $\gamma$ Leporis and $\gamma$ Virginis each in Bands 6 and 7, and $\eta$ Corvi in Band 8, the highest-frequency target searched to date), and four previously single-window entries (GJ 849, HD 285968, HD 23484, and HD 53143) have been expanded to their full set of searched windows. A total of 23 target/bands that completed *processing* are deliberately *excluded* from the table entirely: an automated pointing sanity check (§4) found their assigned ALMA dataset's actual pointing lies tens to hundreds of primary-beam-widths from the star's catalogued position, indicating an upstream target-MOUS crossmatch error rather than a genuine result for that star (four of these were reported as excluded in an earlier version; a further 19 were found, and a related pipeline bookkeeping bug fixed, in the version immediately preceding this one – see §6). A further target, $\gamma$ Lupi (Band 6), is withheld for a different reason: a genuine partial-crossmatch case in which most, but not all, of its spectral windows and epochs are correctly pointed, combined with one surviving window's implausibly deep result from only 12 s of integration, leaves us with insufficient confidence in this target's processing to report it yet (§5). All excluded/withheld target/bands are being re-investigated. This table will continue to grow, and existing rows will be superseded by per-window results, as the survey proceeds; the version current at the time of writing is reproduced here so that the summary statistics quoted in the main text are fully traceable.

## REFERENCES


Backus P. R., Project Phoenix Team, 2002, in Bulletin of the American Astronomical Society. p. 1173
Choza C., et al., 2024, AJ, 168, 254
Cocconi G., Morrison P., 1959, Nature, 184, 844
Czech D., et al., 2021, PASP, 133, 064502
Czech D. J., et al., 2026, arXiv:2607.23651
Drake F. D., 1960, Physics Today, 13, 40
Drake F. D., 1974, in Sagan C., ed., Communication with Extraterrestrial Intelligence (CETI). MIT Press
Drake F. D., 1986, in Applications of Digital Image Processing IX. SPIE
Enriquez E., Price D., 2019, turboSETI, Astrophysics Source Code Library, record ascl:1906.006
Enriquez J. E., et al., 2017, ApJ, 849, 104
Gajjar V., et al., 2019, in Bulletin of the American Astronomical Society. AAS Meeting #233
Garrett M. A., et al., 2026, MNRAS, accepted (arXiv:2605.10212)
Huang B.-L., Tao Z.-Z., Zhang T.-J., 2026, ApJ, 1006, 9
Isaacson H., et al., 2017, PASP, 129, 054501
Jacobson-Bell K., et al., 2025, AJ, 169, 233
Johnson O. A., et al., 2023, AJ, 166, 193
Lacki B. C., et al., 2021, arXiv:2101.02244
Li J.-K., Tao Z.-Z., Zhang T.-J., 2026, arXiv:2603.19023
Ma P. X., et al., 2023, Nature Astronomy, 7, 492
Margot J.-L., et al., 2023, AJ, 166, 189
Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2024, MNRAS, 536, 2127
Morrison I. S., et al., 2023, PASA, 40, e038
Ng C., et al., 2022, AJ, 164, 205
Poznanski D., et al., 2025, AJ, in press (arXiv:2505.03927)
Price D. C., et al., 2020, AJ, 159, 86
Sheikh S. Z., 2020, International Journal of Astrobiology, 19, 237
Sheikh S. Z., et al., 2025, AJ, 169, 108
Sheikh S. Z., et al., 2025, arXiv:2512.18142 ("A Search for Radio Technosignatures from Interstellar Object 3I/ATLAS with the Allen Telescope Array")
Siemion A. P. V., et al., 2015, arXiv:1412.4867
Tarter J., 2001, ARA&A, 39, 511
Tremblay C. D., et al., 2024, PASA, 41, e004
Tremblay C. D., et al., 2026, in Advancing Astrophysics with the SKA II (AASKAII), arXiv:2606.27565
Tusay N., Sheikh S. Z., Sneed E. L., Farah W., Pollak A. W., Cruz L. F., Siemion A., DeBoer D. R., Wright J. T., 2024, AJ, 168, 283
Tusay N., et al., 2025, arXiv:2506.13459
Valente G., et al., 2025, Acta Astronautica, in press
Vidal C., et al., 2026, arXiv:2605.21093 ("The Search for Technosignatures: a Review of Possibilities")
Von Korff J., et al., 2025, AJ, 170, 112
White G. J., 2026, MNRAS, submitted
Wright J. T., Kanodia S., Lubar E. G., 2018, AJ, 156, 260
Zhang Z.-S., et al., 2020, ApJ, 891, 174

TABLE 1
Per-target/band results, ordered by distance. Band numbers are given without the customary "B" prefix to avoid confusion with B-type spectral classifications elsewhere in the table. Frequency range is in GHz; $N_{spw}$ is the total number of spectral windows configured simultaneously in the observation. $EIRP_{min}$ (W) is the minimum EIRP that would register as a $5\sigma$ detection in that spectral window (§4), primary-beam corrected. Flux is the measured peak continuum flux density for a detection, or the resulting $5\sigma$ upper limit (denoted "<"), also primary-beam corrected; RMS is the underlying (post-correction) continuum image rms noise; both in mJy. For a target with more than one searched spectral window, its single continuum measurement (common to all windows in that band) is given once, in the first row. Both automatically-flagged technosignature candidates are noted directly in the Notes column; see §5 for their disposition. This table spans four parts (Tables 1–4) for typesetting reasons only; it is a single table.

| Target | Band | Sp. type | Dist. (pc) | Freq. range (GHz) | $N_{spw}$ | $EIRP_{min}$ (W) | Flux (mJy) | RMS (mJy) | Notes |
|---|---|---|---|---|---|---|---|---|---|
| Proxima Cen | 6 | M6V | 1.30 | 223.09–224.91 | 4 | 3.43e+13 | <0.666 | 0.1332 | reprocessing queued (other spws) |
| Barnard's Star | 7 | M5V | 1.83 | ... | 6 | ... | <3.622 | 0.7243 | narrowband search failed; reprocessing queued |
| Wolf 359 | 7 | M6V | 2.41 | ... | 7 | ... | <3.380 | 0.6761 | narrowband search failed; reprocessing queued |
| Sirius B | 3 | DA2 | 2.67 | 92.13–93.86 | 4 | 2.65e+13 | <0.196 | 0.0393 | primary-beam corrected (×1.036, §4); reprocessing queued (other spws) |
| Sirius B | 4 | DA2 | 2.67 | 142.13–143.86 | 4 | 2.53e+13 | <0.271 | 0.0540 | primary-beam corrected (×1.096); reprocessing queued (other spws) |
| Sirius B | 5 | DA2 | 2.67 | 195.13–196.86 | 4 | 4.04e+13 | <0.528 | 0.1057 | primary-beam corrected (×1.160); reprocessing queued (other spws) |
| G 272-61B | 3 | M6Ve | 2.67 | 89.63–91.36 | 4 | 1.65e+13 | 0.850 | 0.0260 | unresolved UV Cet AB pair, blended flux; reprocessing queued (other spws) |
| G 272-61A | 3 | M5.5Ve | 2.72 | 89.63–91.36 | 4 | 1.71e+13 | 0.850 | 0.0260 | unresolved UV Cet AB pair, blended flux; reprocessing queued (other spws) |
| CD-23 14742 | 6 | M5V | 2.98 | 223.09–224.91 | 4 | 1.20e+14 | <0.388 | 0.0776 | reprocessing queued (other spws) |
| tau Cet | 6 | K0V | 3.65 | 229.55–231.55 | 24 | 5.46e+13 | 0.914 | 0.1323 | reprocessing queued (other spws) |
| HD 33793 | 6 | sdM1 | 3.93 | 223.09–224.91 | 4 | 5.62e+14 | <0.578 | 0.1157 | Kapteyn's Star; reprocessing queued (other spws) |
| Wolf 28 | 6 | DZ7 | 4.31 | 223.13–224.87 | 4 | 6.67e+13 | <0.161 | 0.0321 | Van Maanen's Star (white dwarf); reprocessing queued (other spws) |
| GJ 674 | 6 | M3V | 4.55 | 223.09–224.91 | 4 | 2.74e+14 | <0.516 | 0.1032 | reprocessing queued (other spws) |
| GJ 581 | 6 | M4V | 6.30 | 223.01–224.99 | 4 | 6.35e+14 | <0.451 | 0.0903 | 4/4 spws searched, no candidate |
| GJ 581 | 6 | M4V | 6.30 | 225.01–226.99 | 4 | 6.01e+14 | — | — | |
| GJ 581 | 6 | M4V | 6.30 | 239.01–240.99 | 4 | 6.61e+14 | — | — | |
| GJ 581 | 6 | M4V | 6.30 | 241.01–242.99 | 4 | 6.97e+14 | — | — | |
| 61 Vir | 7 | G5V | 8.53 | 344.87–346.71 | 8 | 4.52e+14 | <1.047 | 0.2094 | 4/8 spws searched |
| 61 Vir | 7 | G5V | 8.53 | 346.82–348.56 | 8 | 2.87e+15 | — | — | |
| 61 Vir | 7 | G5V | 8.53 | 334.82–336.56 | 8 | 2.54e+15 | — | — | |
| 61 Vir | 7 | G5V | 8.53 | 332.93–334.66 | 8 | 3.20e+15 | — | — | |
| GJ 849 | 6 | M4V | 8.81 | 212.09–213.91 | 4 | 4.18e+15 | <0.712 | 0.1424 | |
| GJ 849 | 6 | M4V | 8.81 | 214.09–215.91 | 4 | 4.11e+15 | — | — | |
| GJ 849 | 6 | M4V | 8.81 | 227.09–228.91 | 4 | 4.37e+15 | — | — | |
| GJ 849 | 6 | M4V | 8.81 | 229.08–230.92 | 4 | 8.30e+14 | — | — | |

TABLE 2
Table 1 continued.

| Target | Band | Sp. type | Dist. (pc) | Freq. range (GHz) | $N_{spw}$ | $EIRP_{min}$ (W) | Flux (mJy) | RMS (mJy) | Notes |
|---|---|---|---|---|---|---|---|---|---|
| gam Lep | 6 | F7V | 8.91 | 223.09–224.91 | 4 | 1.02e+14 | 0.349 | 0.0165 | |
| gam Lep | 6 | F7V | 8.91 | 225.09–226.91 | 4 | 9.27e+13 | — | — | |
| gam Lep | 6 | F7V | 8.91 | 239.09–240.91 | 4 | 1.02e+14 | — | — | |
| gam Lep | 6 | F7V | 8.91 | 241.09–242.91 | 4 | 1.09e+14 | — | — | |
| gam Lep | 7 | F7V | 8.91 | 335.63–337.36 | 4 | 2.86e+14 | 0.695 | 0.0623 | |
| gam Lep | 7 | F7V | 8.91 | 337.56–339.30 | 4 | 2.36e+14 | — | — | |
| gam Lep | 7 | F7V | 8.91 | 347.63–349.36 | 4 | 2.65e+14 | — | — | |
| gam Lep | 7 | F7V | 8.91 | 349.63–351.36 | 4 | 3.14e+14 | — | — | |
| HD 285968 | 6 | M1V | 9.49 | 212.05–213.87 | 4 | 1.93e+15 | <0.700 | 0.1400 | |
| HD 285968 | 6 | M1V | 9.49 | 214.05–215.87 | 4 | 1.83e+15 | — | — | |
| HD 285968 | 6 | M1V | 9.49 | 227.04–228.87 | 4 | 1.97e+15 | — | — | |
| HD 285968 | 6 | M1V | 9.49 | 229.04–230.88 | 4 | 3.45e+14 | — | — | |
| AU Mic | 6 | M1Ve | 9.71 | 229.68–231.40 | 4 | 5.26e+13 | 0.633 | 0.0374 | automated candidate flag, see text |
| AU Mic | 6 | M1Ve | 9.71 | 227.63–229.36 | 4 | 2.64e+14 | — | — | |
| AU Mic | 6 | M1Ve | 9.71 | 212.63–214.36 | 4 | 2.78e+14 | — | — | |
| AU Mic | 6 | M1Ve | 9.71 | 215.13–216.86 | 4 | 2.95e+14 | — | — | |
| AT Mic B | 7 | M4V | 9.81 | 347.07–348.80 | 4 | 3.43e+14 | 0.337 | 0.0292 | |
| AT Mic B | 7 | M4V | 9.81 | 335.07–336.80 | 4 | 2.88e+14 | — | — | |
| AT Mic B | 7 | M4V | 9.81 | 333.17–334.91 | 4 | 3.30e+14 | — | — | |
| AT Mic B | 7 | M4V | 9.81 | 345.18–346.90 | 4 | 5.60e+13 | — | — | |
| AT Mic AB | 7 | M5V | 9.92 | 347.07–348.80 | 4 | 3.56e+14 | 0.344 | 0.0300 | |
| AT Mic AB | 7 | M5V | 9.92 | 335.07–336.80 | 4 | 3.00e+14 | — | — | |
| AT Mic AB | 7 | M5V | 9.92 | 333.17–334.91 | 4 | 3.44e+14 | — | — | |
| AT Mic AB | 7 | M5V | 9.92 | 345.18–346.90 | 4 | 5.82e+13 | — | — | |

TABLE 3
Table 1 continued.

| Target | Band | Sp. type | Dist. (pc) | Freq. range (GHz) | $N_{spw}$ | $EIRP_{min}$ (W) | Flux (mJy) | RMS (mJy) | Notes |
|---|---|---|---|---|---|---|---|---|---|
| gam Vir | 6 | F0V | 12.02 | 223.09–224.91 | 4 | 2.49e+14 | 0.326 | 0.0135 | |
| gam Vir | 6 | F0V | 12.02 | 225.09–226.91 | 4 | 2.24e+14 | — | — | |
| gam Vir | 6 | F0V | 12.02 | 239.09–240.91 | 4 | 2.49e+14 | — | — | |
| gam Vir | 6 | F0V | 12.02 | 241.09–242.91 | 4 | 2.59e+14 | — | — | |
| gam Vir | 7 | F0V | 12.02 | 335.58–337.41 | 4 | 4.43e+14 | 0.648 | 0.0244 | |
| gam Vir | 7 | F0V | 12.02 | 337.52–339.35 | 4 | 3.86e+14 | — | — | |
| gam Vir | 7 | F0V | 12.02 | 347.58–349.41 | 4 | 4.18e+14 | — | — | |
| gam Vir | 7 | F0V | 12.02 | 349.58–351.41 | 4 | 4.73e+14 | — | — | |
| HR 1010 | 6 | G2V | 12.04 | 229.61–231.45 | 4 | 6.50e+14 | <0.498 | 0.0997 | |
| HR 1010 | 6 | G2V | 12.04 | 231.62–233.35 | 4 | 1.75e+15 | — | — | |
| HR 1010 | 6 | G2V | 12.04 | 244.62–246.35 | 4 | 1.87e+15 | — | — | |
| HR 1010 | 6 | G2V | 12.04 | 246.62–248.35 | 4 | 2.21e+15 | — | — | |
| HR 1010 | 7 | G2V | 12.04 | 345.32–346.24 | 4 | 1.27e+15 | <1.353 | 0.2706 | |
| HR 1010 | 7 | G2V | 12.04 | 347.05–348.87 | 4 | 6.34e+15 | — | — | |
| HR 1010 | 7 | G2V | 12.04 | 333.09–334.92 | 4 | 6.09e+15 | — | — | |
| HR 1010 | 7 | G2V | 12.04 | 335.05–336.87 | 4 | 5.33e+15 | — | — | |
| TRAPPIST-1 | 3 | M8V | 12.47 | 89.63–91.36 | 4 | 1.57e+14 | <0.036 | 0.0071 | |
| TRAPPIST-1 | 3 | M8V | 12.47 | 91.56–93.30 | 4 | 1.55e+14 | — | — | |
| TRAPPIST-1 | 3 | M8V | 12.47 | 101.63–103.36 | 4 | 1.57e+14 | — | — | |
| TRAPPIST-1 | 3 | M8V | 12.47 | 103.63–105.36 | 4 | 1.59e+14 | — | — | |
| TRAPPIST-1 | 6 | M8V | 12.47 | 223.08–224.92 | 4 | 5.08e+15 | <0.531 | 0.1061 | |
| TRAPPIST-1 | 6 | M8V | 12.47 | 225.08–226.92 | 4 | 4.72e+15 | — | — | |
| TRAPPIST-1 | 6 | M8V | 12.47 | 239.08–240.92 | 4 | 5.19e+15 | — | — | |
| TRAPPIST-1 | 6 | M8V | 12.47 | 241.08–242.92 | 4 | 5.49e+15 | — | — | |
| TRAPPIST-1 | 7 | M8V | 12.47 | 347.62–349.36 | 4 | 3.47e+14 | <0.106 | 0.0212 | |
| TRAPPIST-1 | 7 | M8V | 12.47 | 335.62–337.36 | 4 | 2.86e+14 | — | — | |
| TRAPPIST-1 | 7 | M8V | 12.47 | 333.73–335.46 | 4 | 3.31e+14 | — | — | |
| TRAPPIST-1 | 7 | M8V | 12.47 | 345.73–347.46 | 4 | 5.76e+13 | — | — | |

TABLE 4
Table 1 continued.

| Target | Band | Sp. type | Dist. (pc) | Freq. range (GHz) | $N_{spw}$ | $EIRP_{min}$ (W) | Flux (mJy) | RMS (mJy) | Notes |
|---|---|---|---|---|---|---|---|---|---|
| HD 38858 | 6 | G5V | 15.21 | 244.09–245.82 | 4 | 3.56e+15 | <0.686 | 0.1373 | |
| HD 38858 | 6 | G5V | 15.21 | 246.08–247.82 | 4 | 3.97e+15 | — | — | |
| HD 38858 | 6 | G5V | 15.21 | 231.59–233.32 | 4 | 3.18e+15 | — | — | |
| HD 38858 | 6 | G5V | 15.21 | 229.64–231.36 | 4 | 6.96e+14 | — | — | |
| HD 207129 | 6 | G2V | 15.56 | 244.15–245.88 | 4 | 4.03e+15 | <0.563 | 0.1126 | |
| HD 207129 | 6 | G2V | 15.56 | 246.15–247.88 | 4 | 5.06e+15 | — | — | |
| HD 207129 | 6 | G2V | 15.56 | 229.64–231.48 | 4 | 7.62e+14 | — | — | |
| HD 207129 | 6 | G2V | 15.56 | 231.65–233.38 | 4 | 3.89e+15 | — | — | |
| HD 23484 | 6 | K5V | 16.17 | 229.61–231.45 | 4 | 1.70e+15 | <0.497 | 0.0995 | |
| HD 23484 | 6 | K5V | 16.17 | 214.57–216.40 | 4 | 5.12e+15 | — | — | |
| HD 23484 | 6 | K5V | 16.17 | 216.57–218.40 | 4 | 5.02e+15 | — | — | |
| HD 23484 | 6 | K5V | 16.17 | 231.57–233.40 | 4 | 5.36e+15 | — | — | |
| HD 10647 | 7 | G0V | 17.35 | 330.35–332.19 | 4 | 1.97e+15 | <2.020 | 0.4040 | reprocessing queued (other spws) |
| eta Crv | 8 | F2V | 18.24 | 479.34–481.17 | 4 | 6.08e+16 | <3.609 | 0.7218 | |
| eta Crv | 8 | F2V | 18.24 | 481.23–483.06 | 4 | 6.13e+16 | — | — | |
| eta Crv | 8 | F2V | 18.24 | 493.30–495.13 | 4 | 7.10e+16 | — | — | |
| eta Crv | 8 | F2V | 18.24 | 492.19–492.24 | 4 | 3.16e+15 | — | — | |
| HD 53143 | 6 | K0V | 18.34 | 214.57–216.39 | 4 | 8.45e+15 | <0.877 | 0.1754 | 4/4 spws searched, no candidate |
| HD 53143 | 6 | K0V | 18.34 | 216.57–218.39 | 4 | 8.49e+15 | — | — | |
| HD 53143 | 6 | K0V | 18.34 | 229.60–231.44 | 4 | 2.84e+15 | — | — | |
| HD 53143 | 6 | K0V | 18.34 | 231.57–233.39 | 4 | 8.97e+15 | — | — | |
| bet Pic | 3 | F0V | 19.63 | 114.83–115.69 | 4 | 2.08e+14 | 0.444 | 0.0281 | CO(1-0) line, candidate; see text |
| bet Pic | 6 | F0V | 19.63 | 230.08–231.00 | 4 | 2.01e+15 | 5.659 | 0.9626 | reprocessing queued (other spws) |